THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

# MATH2014

# Multivariable Calculus and Linear Algebra

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# Chapter One

1. Determine whether the function

$$f(x,y) = \begin{cases} \dfrac{\sqrt{xy+1}-1}{xy} & x \neq 0 \text{ and y} \neq 0 \text{ and xy} \geq -1, \\ 0 & (x,y) = (0,0). \end{cases}$$

is continuous at $(0,0)$ or not. Prove your result.

2. Determine whether the function

$$f(x,y) = \begin{cases} \dfrac{2x^2y}{x^4+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0). \end{cases}$$

is continuous at $(0,0)$ or not. Prove your result.

3. Find the absolute minimum and absolute maximum value of the function $f(x,y) = \sqrt{x^2+y^2-2x+2}$ on the closed half disk $R = \{(x,y) : x^2+y^2 \leq 4, y \geq 0\}$.

4. Find the absolute minimum and absolute maximum value of the function for $f(x,y) = x^2y(4-x-y)$ in the closed region bounded by $x-$axis, $y-$axis and the line $x+y=6$.

5. The temperature at each point of the sphere $x^2+y^2+z^2 = 11$ is given by the function $T(x,y,z) = 20+2x+2y+z^2$. Find the highest and lowest temperature on the intersection of the given sphere $x^2+y^2+z^2 = 11$ and the plane $x+y+z=3$.

6. Using ONE iteration of Newton-Raphson's method with initial guess $(1,-1)$ to solve the system of linear equations

$$\begin{cases} x^2+4y^2-9=0 \\ 18y-14x^2+45=0 \end{cases}.$$

7. Find all critical points of the function $w = x^3+y^3-12x-3y+20$. Also determine the nature of the critical points (relative maximum, relative minimum or saddle point).

8. Find all absolute maxima and minima of the function $f(x,y) = -x^2-y^2+2x+4y+2$ on the closed first quadrant $x \geq 0$, $y \geq 0$.

9. Let $w = x^2e^{2y}\cos(3z)$. Find the value of $dw/dt$ at the point $(1, \ln 2, 0)$ on the curve $x = \cos t$, $y = \ln(t+2)$, $z = t$.

10. Find the critical point of $f(x,y) = xy + 2x - \ln(x^2 y)$ in the open first quadrant $x > 0$ and $y > 0$. Also determine the nature of this critical point (relative maximum, relative minimum or saddle point).

11. Find and classify all the critical points for $f(x,y) = 3x^2y + y^3 - 3x^2 - 3y^2 + 2$.

12. Determine whether the function

$$f(x,y) = \begin{cases} \frac{x^2y}{x^2+y^2} & (x,y) \neq (0,0), \\ 0 & (x,y) = (0,0). \end{cases}$$

is continuous at $(0,0)$ or not. Prove your result.

13. Find the absolute minimum and absolute maximum value of the function for $f(x,y) = x^4 + y^4 - 4xy + 2$ on the region $D = \{(x,y) : 0 \leq x \leq 3, 0 \leq y \leq 2\}$

14. Use Lagrange multipliers to find the minimum and maximum value of $f(x,y,z) = xyz$ on the surface $x^2 + 2y^2 + 3z^2 = 6$.

15. Using ONE iteration of Newton-Raphson's method, estimate an intersection point of the circle $x^2 + y^2 = 1$ and the parabola $y = x^2$ with the initial guess $(1,1)$.

16. Let $f(x,y) = 2x^2y^2 + x^2y - 2x - y^2$. Using ONE iteration of Newton-Raphson's method, estimate one critical point of $f$ with the initial guess $(1,-1)$. (Round your answers to 4 decimal places.)

17. Find the shortest distance from the the origin to intersection of $z^2 = x^2 + y^2$ and $x - 2z = 3$.

18. Find the shortest distance from the point $P = (-1,2,3)$ to the line of intersection of the planes $x + 2y - 3z = 4$ and $2x - y + 2z = 5$.

19. Suppose you are running a company, producing some sort of widget that requires steel as a raw material. Your costs consist of human labor, which is

\$20 per hour, and the steel itself, which runs for \$170 per ton. Suppose your revenue is modeled by the following equation:

$$R(h, s) = 200h^{\frac{2}{3}}s^{\frac{1}{3}},$$

where $h$ represents hours of labor and $s$ represents tons of steel. If your budget is \$20000, what is the maximum possible revenue? (Round your answers to the nearest dollar.)

20. The specific gravity $S$ of a solid may be calculated by the formula $S = \frac{x}{x-y}$, where $x$ is the weight (in *Newtons*) of the solid in air and $y$ is the weight (in *Newtons*) of the solid in water. Suppose that in a certain experiment, $x$ is measured to be 9 with an error not exceeding 0.05 and $y$ is measured to be 5 with an error not exceeding 0.1. If $\Delta S$ denotes the error in calculating $S$ by these measurements, use total differential to estimate the maximum of $\Delta S$.(Round your answers to 2 decimal places.)

21. Let $f(x, y) = -x^2y + y^2 + 2y$. Using ONE iteration of Newton-Raphson's method, estimate the minimum of $f$ with the initial guess $(-3, -3)$. (Round your answers to 2 decimal places.)

22. Using ONE iteration of Newton- Raphson's method, find the first approximate for the solution to the system of equations

$$\begin{cases} x^2 + y^2 = 4 \\ y + e^x = 1 \end{cases}$$

with initial guess $(1, -1.7)$. (Round your answers to 4 decimal places.)

23. Using ONE iteration of Newton- Raphson's method, solve the system of equations

$$\begin{cases} x^2 + x - y^2 - 1 = 0 \\ y - \sin(x^2) = 0 \end{cases}$$

with initial guess $(0.7, 0.5)$ (Round your answers to 4 decimal places.)

24. Using ONE iteration of Newton-Raphson's method with initial guess $(0.5, 1.5)$ to find an approximation to the critical point of $f(x, y) = x^x \sin y$. (Round your solution up to 4 decimal places.)

25. Use the method of Lagrange multiplier to find the area of the intersecting surface of $x^2 + y^2 + 4z^2 = 1$ and $x + y + z = 0$. (Hint: the area of an ellipse equals to $\pi ab$, where $a$ and $b$ are the semi-axes of the ellipse.)

26. It is known that the temperature $T$ at any point $(x, y, z)$ is given by $T = 2x^2 + 5y^2 + 11z^2 + 20xy - 4xz + 16yz$. Find the hottest point(s) on the unit sphere $x^2 + y^2 + z^2 = 1$.

27. Given that $z = \sin y + f(\sin x - \sin y)$, where $f$ is a differentiable function. Simplify $\sec x \dfrac{\partial z}{\partial x} + \sec y \dfrac{\partial z}{\partial y}$.

28. Determine whether the function

$$f(x, y) = \begin{cases} \dfrac{x^2y^2}{x^2y^2 + (x-y)^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$ or not. Prove your result.

29. Find all critical points of the function $f(x, y) = (y - x^2)(y - x^4)$. Also determine the nature of the critical points (relative maximum, relative minimum or saddle point).

30. Find the absolute minimum and absolute maximum value of the function $f(x, y, z) = x^2 - y^2$ on the surface $x^2 + 2y^2 + 3z^2 = 1$.

31. Determine whether the function

$$f(x, y) = \begin{cases} \dfrac{x^4y^4}{(x^2 + y^4)^3} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$ or not. Prove your result.

32. Calculate $\dfrac{\partial f}{\partial x}$ where $f(x, y) = (x + 2y)^{(2y-3x)}$.

# Chapter Two

33. Find the area inside the cardioid $r = 1+\sin\theta$ and outside the circle $r = 3\sin\theta$.

34. Evaluate $\iint_R \frac{y}{x}\,dx\,dy$, where $R$ is the region in the $xy$-plane bounded by the curves $x^2 - y^2 = 1$, $x^2 - y^2 = 4$, $y = 0$ and $y = \frac{x}{2}$. (Keep EXACT solution.)

35. Given that the density at any point is proportional to its distance from the center of the base, find the mass of a solid hemisphere $H$ with radius $a$.

    (a) Write the integral in cylindrical coordinates.

    (b) Write the integral in spherical coordinates.

    (c) Evaluate any ONE of the integral in part (a) and (b).

36. Evaluate $\iint_R (x^2 - xy - 2y^2)\,dx\,dy$, where $R$ is the region in the $xy$-plane bounded by the lines $x - 2y = 0$, $x - 2y = 1$, $x + y = 0$ and $x + y = 1$.

37. Find the volume of the solid region $A$ between $x^2+y^2+z^2 = a^2$ and $x^2+y^2 = ax$ $(a > 0)$.

38. Find the volume of the solid region $B$ in the first octant between $x+y+z^2 = 1$, $x = 0$, $y = 0$ and $z = 0$.

39. Evaluate the double integral $\int\int_\Omega (x+y)dxdy$, where $\Omega$ is the region between $y = x^3$ and $y = x^4$, $x \in [-1, 1]$.

40. Evaluate the double integral $\int\int_D dxdy$, where $D$ is the region in the first quadrant bounded by $xy = 4$, $xy = 8$, $xy^3 = 5$ and $xy^3 = 15$.

41. Evaluate $\int\int_R \left(\frac{x-y}{x+y+2}\right)^2 dxdy$ over the region $R$ below.

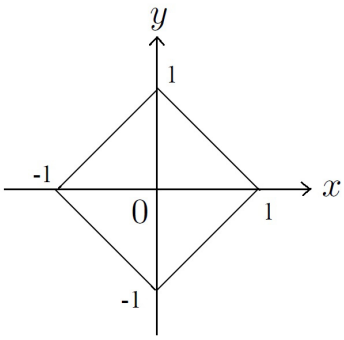



Figure 1: Region $R$

42. Find the volume of the solid that lies above the cone $z = \sqrt{x^2+y^2}$ and below $x^2+y^2+z^2 = z$.

43. Compute $\int_{-1}^{1}\int_{0}^{\sqrt{1-x^2}}\int_{(x^2+y^2)^3}^{5} \sqrt{x^2+y^2}dzdydx$.

44. Find the area of the region bounded by the curves $r = \cos\theta$ and $r = \sin\theta$.

45. Find the volume of the solid region $E$ between $y = 4-x^2-z^2$ and $y = x^2+z^2$.

46. Evaluate $\iiint_E zdV$ where $E$ is the portion of the solid sphere $x^2+y^2+z^2 \leq 9$ that is inside the cylinder $x^2+y^2 = 1$ and above the cone $x^2+y^2 = z^2$.

47. Evaluate the integral $\iint_R \cos\left(\frac{x-y}{x+y}\right) dxdy$, where $R$ is the triangular region with vertices (0, 0), (1, 0), (0, 1). (Keep the exact answer. Do NOT round your answer.)

48. Evaluate the volume of D where D is the solid right cylinder whose base is the region between the circles $r = \cos\theta$ and $r = 2\cos\theta$ and whose top lies in the plane $z = 3-y$.

49. Find the volume of the solid bounded by the hyperboloid $z^2 = a^2+x^2+y^2$ and the cone $z = \sqrt{2(x^2+y^2)}$.

50. Find the mass of a ball of radius $R$ given that the density of each point equals the distance from this point to the boundary.

51. Find the centroid ($\delta = 1$) of the solid enclosed by the cylinder $x^2 + y^2 = 4$, bounded above by the paraboloid $z = x^2 + y^2$, and bounded below by the $xy-$plane.

52. Use spherical coordinate to evaluate

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{0}^{\sqrt{4-x^2-y^2}} z^2 \sqrt{x^2 + y^2 + z^2} dz dy dx.$$

53. Evaluate the integral

$$\int_0^2 \int_{y/2}^{(y+4)/2} (2x - y) y^3 e^{(2x-y)^2} dx dy.$$

(Hint: You can use the transformation: $x = u + v/2$ and $y = v$.)

54. Find the volume of the solid which is bounded below by $z = \sqrt{3x^2 + 3y^2}$ and bounded above by $x^2 + y^2 + z^2 = 16$.

55. Find the mass of a circular cone of base radius $r = 1$ and height $h = 2$ given that the density equals the distance from the vertex.

56. Evaluate the double integral $\iint_R x dx dy$, where $R$ is region inside the circle $(x-1)^2 + y^2 = 1$ and outside the circle $(x-2)^2 + y^2 = 1$.

57. Find the volume of material cut from the solid sphere $r^2 + z^2 \leq 9$ by the cylinder $r = 3 \sin \theta$.

58. Find the volume of the region enclosed by the surfaces $x+y+z = a$, $x^2+y^2 = R^2$, $x = 0$, $y = 0$, $z = 0$, where $a \geq \sqrt{2}R$.

59. Evaluate $\displaystyle\int_0^1 \frac{f(x)}{\sqrt{x}} dx$, where $f(x) = \displaystyle\int_1^x \frac{\ln(1+t)}{t} dt$. (Keep EXACT solution. )

60. Find the absolute minimum and absolute maximum value of the function $f(x, y) = \sin x + \sin y - \sin(x + y)$ on the closed region $D = \{(x, y) : x \geq 0, y \geq 0, x + y \leq 2\pi\}$.

61. Find the area bounded by the parabolas $y^2 = mx$, $y^2 = nx$ $(m > n > 0)$ and the lines $y = px$, $y = qx$ $(p > q > 0)$.

62. Evaluate the double integral $\iint_R y dx dy$, where $R$ is region inside the circle $x^2 + (y-1)^2 = 1$ and outside the circle $x^2 + (y-2)^2 = 1$.

63. Find the volume of the region enclosed by the sphere $x^2 + y^2 + z^2 = R^2$ and the cylinder $x^2 + y^2 = Rx$ $(R > 0)$.

64. A flat circular plate has the shape of a circular disc $x^2 + y^2 \leq 1$. The temperature $T$ at any point $(x, y)$ on the plate, including the boundary where $x^2 + y^2 = 1$, is given by $T = x^2 + y^2 - x - y + xy$. Find the hottest and coldest points on the plate and temperature at each of these points.

# Chapter Three

65. Give answers only to the following questions. The process of calculating is **NOT** required.

(a) Find the value(s) of $h$ for which the following set of vectors $\{v_1 = (1,0,0)^T, v_2 = (h,1,-h)^T, v_3 = (1,2h,3h+1)^T\}$ is linearly independent.

(b) For which choice(s) of the constant $k$ is the following matrix invertible?

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & k \\ 1 & 4 & k^2 \end{pmatrix}.$$

(c) $A$ is an $n \times n$ matrix. $|A| = x$, the rank of $A$ is $y$ and $p$ is one of the eigenvalues of $A$. If we double every entry of $A$ and get a new matrix $B$, find

(i) the determinant of $B$.

(ii) the rank of $B$.

(iii) one eigenvalue of $B$.

66. For each of the following statements, write (T) if it is true and (F) if it is false, and then give a brief (say, one-line) explanation. Each question carries 2.5 marks.

(a) A linear system of four equations in three unknowns is always inconsistent.

(b) A linear system with fewer equations than unknowns must have infinitely many solutions.

(c) If the system $Ax = b$ has a unique solution, then $A$ must be a square matrix.

(d) A homogeneous system is always consistent.

67. Given that

$$A = \begin{bmatrix} 5 & 4 & 2 \\ 4 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of $A$ (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix $P$ and a diagonal matrix $D$, such that $A = PDP^{-1}$.

(c) Find the $(3,3)$-entry of $A^{2018}$.

68. Solve the system of linear equations

$$\begin{cases} x_1 - 2x_2 + 3x_3 = 6 \\ 2x_1 - 3x_2 + 2x_3 = 2 \\ x_1 + 2x_2 - 5x_3 = -10 \end{cases}$$

by

(a) Inverse Matrix method.

(b) Cramer's rule.

69. Find all values of $\lambda$ for which the homogeneous system of linear equations has nontrivial solutions.

$$\begin{cases} (\lambda+2)x_1 - 2x_2 + 3x_3 = 0 \\ -2x_1 + (\lambda-1)x_2 + 6x_3 = 0 \\ x_1 + 2x_2 + \lambda x_3 = 0 \end{cases}$$

70. (a) Consider a linear system whose augmented matrix is of the form

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & 1 & \beta & 0 \end{array}\right).$$

i. Is it possible for the system to be inconsistent? Explain.

ii. For what values of $\beta$ will the system have infinitely many solutions?

(b) Consider a linear system whose augmented matrix is of the form

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right).$$

i. For what values of $a$ and $b$ will the system have infinitely many solutions?

ii. For what values of $a$ and $b$ will the system be inconsistent?

iii. For what values of $a$ and $b$ will the system have unique solution?

71. Given that

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of $A$ (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix $P$ and a diagonal matrix $D$, such that $A = PDP^{-1}$.

(c) Hence, find the $(2;3)-$ entry of $A^{2017}$.

72. A square matrix $A$ is said to be *nilpotent* of index $k$ if there exist a smallest positive integer $k$ such that $A^k = 0$.

(a) What is the index of the nilpotent matrix $\begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$?

(b) If $A$ is nilpotent, what can you say about $A^T$? Prove your answer.

(c) Are nilpotent matrices invertible? Prove your answer.

(d) If $A$ is nilpotent, is $I - A$ invertible? Prove your answer.

73. Given that

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}.$$

(a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix $P$ such that $P^{-1}AP$ is a diagonal matrix.

(c) Hence, evaluate $A^{2016}$.

74. Solve the system of linear equations

$$\begin{cases} x_1 + 2x_2 + 3x_3 = -1 \\ 3x_1 + 8x_2 + 11x_3 = 1 \\ x_1 + 5x_2 + 7x_3 = 1 \end{cases}$$

by

(a) Inverse Matrix method.

(b) Cramer's rule.

75. Solve the system of linear equations

$$\begin{cases} x_1 + 3x_2 + 2x_3 = 2 \\ 3x_1 - 5x_2 - 3x_3 = 2 \\ x_1 + x_2 + 7x_3 = 14 \end{cases}$$

by

(a) Inverse Matrix method.

(b) Cramer's rule.

76. Consider $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & 5 \\ 5 & 8 & -5 & 12 \\ 3 & 16 & -11 & 28 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

(a) Determine conditions on $b_1$, $b_2$ and $b_3$ such that the system of linear equations is consistent.

(b) Solve the linear system when $\mathbf{b} = \begin{bmatrix} 1 & 4 & 4 \end{bmatrix}^T$.

77. Given that $\lambda = 1$, $\lambda = 4$ and $\lambda = 6$ are the three eigenvalues of the matrix

$$A = \begin{bmatrix} 4 & 2 & 0 & 4 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 3 & 3 \\ 0 & 4 & 0 & 7 \end{bmatrix}.$$

Find the fourth eigenvalue of $A$ and an eigenvector associated with it (Let the first non-zero entry of the eigenvector to be 1).

78. Define $a_1 = 1$, $a_2 = 1$ and $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$.

(a) Find a $2 \times 2$ matrix $A$ such that $\begin{bmatrix} a_n \\ a_{n-1} \end{bmatrix} = A \begin{bmatrix} a_{n-1} \\ a_{n-2} \end{bmatrix}$.

(b) Find a nonsingular matrix $P$ such that $P^{-1}AP$ is a diagonal matrix.

(c) Hence, evaluate $A^{2016}$.

79. Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

(a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a nonsingular matrix $P$ such that $P^{-1}AP$ is a diagonal matrix.

(c) Hence, evaluate $A^{2015}$.

80. Solve the system of linear equations

$$\begin{cases} 2x_1 - 2x_2 + 3x_3 = 10 \\ 3x_1 - 3x_2 + 2x_3 = 5 \\ 4x_1 + 2x_2 - 5x_3 = -6 \end{cases}$$

by

(a) Inverse Matrix method.

(b) Cramer's rule.

81. Given that

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -1 & 1 & 0 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of $A$ (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix $P$ such that $P^{-1}AP$ is a diagonal matrix.

(c) Hence, evaluate $A^{2017}$.

82. Read the following information and answer the question.

A cryptogram is a message written according to a secret code (the Greek word kryptos means "hidden"). This question describes a method of using matrix

multiplication to encode and decode messages.

Begin by assigning a number to each letter in the alphabet (with 0 assigned to a blank space), as follows.

$$\begin{array}{cccccc} 0 = _ & 1 = A & 2 = B & 3 = C & 4 = D & 5 = E \\ 6 = F & 7 = G & 8 = H & 9 = I & 10 = J & 11 = K \\ 12 = L & 13 = M & 14 = N & 15 = O & 16 = P & 17 = Q \\ 18 = R & 19 = S & 20 = T & 21 = U & 22 = V & 23 = W \\ 24 = X & 25 = Y & 26 = Z & & & \end{array}$$

Then the message is converted to numbers and partitioned into uncoded row matrices, each having $n$ entries.

Example 1: Write the uncoded row matrices of size $1 \times 3$ for the message *MEET ME MONDAY*.

The coded row matrices are obtained by multiplying each of the uncoded row matrices found in Example 1 by the matrix $A$, as follows.

$$\begin{array}{ccc} \textit{Uncoded Row Matrix} & \textit{Encoding Matrix A} & \textit{Coded Row Matrix} \\ [13 \quad 5 \quad 5] & \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} & = [13 \quad -26 \quad 21] \\ [20 \quad 0 \quad 13] & \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} & = [33 \quad -53 \quad -12] \\ [5 \quad 0 \quad 13] & \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} & = [18 \quad -23 \quad -42] \\ [15 \quad 14 \quad 4] & \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} & = [5 \quad -20 \quad 56] \\ [1 \quad 25 \quad 0] & \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix} & = [-24 \quad 23 \quad 77] \end{array}$$

The sequence of coded row matrices is

$$[13 \quad -26 \quad 21][33 \quad -53 \quad -12][18 \quad -23 \quad -42][5 \quad -20 \quad 56][-24 \quad 23 \quad 77].$$

Finally, removing the brackets produces the cryptogram below.

$$13 \quad -26 \quad 21 \quad 33 \quad -53 \quad -12 \quad 18 \quad -23 \quad -42 \quad 5 \quad -20 \quad 56 \quad -24 \quad 23 \quad 77$$

Example 2: Use the matrix

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -1 & 1 & 3 \\ 1 & -1 & -4 \end{bmatrix}$$

to encode the message MEET ME MONDAY.

Partitioning the message (including blank spaces, but ignoring punctuation) into groups of three produces the following uncoded row matrices.

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| [13 | 5 | 5] | [20 | 0 | 13] | [5 | 0 | 13] | [15 | 14 | 4] | [1 | 25 | 0] |
| *M* | *E* | *E* | *T* | __ | *M* | *E* | __ | *M* | *O* | *N* | *D* | *A* | *Y* | __ |

Note that a blank space is used to fill out the last uncoded row matrix.

On the other hand, we can use the inverse of $A$ to decoding the cryptogram.

Question: Please use the matrix:

$$A = \begin{bmatrix} 3 & -4 & 2 \\ 0 & 2 & 1 \\ 4 & -5 & 3 \end{bmatrix}$$

to decoding the cryptogram: 112 -140 83 19 -25 13 72 -76 61 95 -118 71 20 21 38 35 -23 36 42 -48 32.

83. Given that

$$A = \begin{bmatrix} -1 & -2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of $A$ (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix $P$ and a diagonal matrix $D$, such that $A = PDP^{-1}$.

(c) Find $A^{2018}$.

84. Read the following information and answer the questions.

Many types of applications involve a finite set of states $\{S_1, S_2, \cdots \ldots S_n\}$ of a given population. For instance, residents of Hong Kong may live in Hong Kong Island, Kowloon or New Territories. The probability that a member of a population will change from the $j-$th state to the $i-$th state is represent by a number $p_{ij}$, where $0 < p_{ij} < 1$. A probability of $p_{ij} = 0$ means that the member is certain not to change from the $j-$th state to the $i-$th state. $P$ is called the matrix of transition probabilities because it gives the probabilities of each possible type of transition (or change) within the population.

$$P = \begin{array}{c} \overbrace{\begin{array}{cccc} S_1 & S_2 & \cdots & S_n \end{array}}^{From} \\ \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nn} \end{bmatrix} \end{array} \begin{array}{c} S_1 \\ S_2 \\ \vdots \\ S_n \end{array} \; To$$

At each transition, each member in a given state must either stay in that state or change to another state. For probabilities, this means that the sum of the entries in any column of $P$ is 1. In general, such a matrix is called "stochastic". That is, an $n \times n$ matrix $P$ is called a stochastic matrix if each entry is a number bewteen 0 and 1 and each column of $P$ adds up to 1.

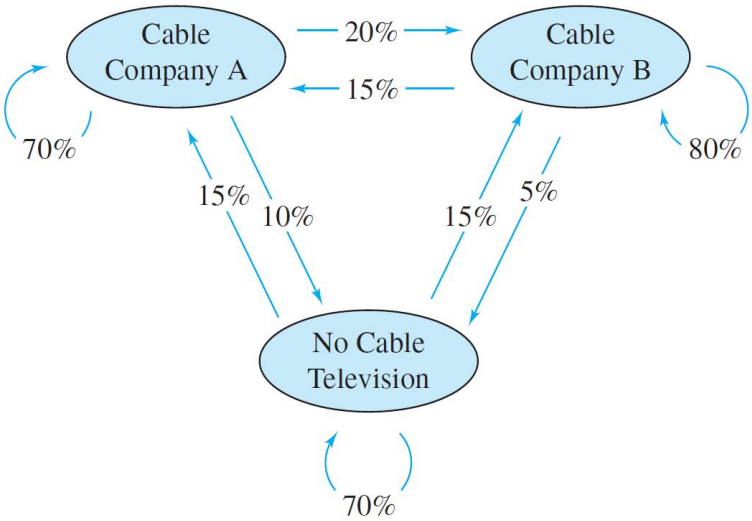
(a) Determine whether the following matrices are stochastic matrices.

$$A = \begin{bmatrix} 0.1 & 0.2 & 0.3 \\ 0.2 & 0.3 & 0.4 \\ 0.3 & 0.4 & 0.5 \end{bmatrix}$$

$$B = \begin{bmatrix} 1/2 & 1/3 & 1/4 \\ 1/4 & 0 & 3/4 \\ 1/4 & 2/3 & 0 \end{bmatrix}$$

(b) Two competing companies in Hong Kong offer cable television service to 100,000 households. The changes in cable subscriptions each year are

shown in the following figure. Company A now has 15,000 subscribers and Company B has 20,000 subscribers. Write the matrix $P$ of transition probabilities and determine how many subscribers will each company have 1 year from now?




- Compute the determinant the following $n \times n$ matrices. Indicate clearly the method being used.

$$A_n = \begin{bmatrix} 1-n & 1 & 1 & \dots & 1 \\ 1 & 1-n & 1 & \dots & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & \dots & 1-n \end{bmatrix}$$

$$B_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & 1-x & 1 & \dots & 1 \\ 1 & 1 & 2-x & \dots & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 1 & 1 & \dots & n-1-x \end{bmatrix}$$

- Let $v_1 = (2,4,3)^T$, $v_2 = (1,3,2)^T$, $v_3 = (5,1,3)^T$.

  (a) Dertermine whether the set $\{v_1, v_2, v_3\}$ is linearly dependent or not. Prove your result.

  (b) If possible, find a linear dependence relation among $v_1, v_2, v_3$.

85. Given that

$$A = \begin{bmatrix} 5 & 8 & 16 \\ 4 & 1 & 8 \\ -4 & -4 & -11 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of $A$ (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix $P$ and a diagonal matrix $D$, such that $A = PDP^{-1}$.

(c) Find the $(3, 3)$-entry of $A^{2019}$.

86. Given that

$$A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$$

(a) Find all the eigenvalues and associated eigenvectors of $A$ (Let the first non-zero entry of each eigenvector to be 1).

(b) Find a non-singular matrix $P$ and a diagonal matrix $D$, such that $A = PDP^{-1}$.

(c) Find the $(3, 3)$-entry of $A^{2019}$.